

Lecture Pre-read materials

This lecture assumes a basic familiarity with Machine Learning, Neural Networks, and sequence modeling. The focus will be on the transition from count-based methods to neural embeddings and representation learning. Please go through the following resources in advance.

Essential Pre-reads —

- Short Notes — A concise reference covering the key concepts for this lecture: [View Notes](#)
- Andrew Ng — Introduction to Deep Learning — Supervised/unsupervised learning, cost functions, and gradient descent: [Watch on YouTube](#)
- Stanford CS224d Lecture Notes — Word meaning, word vectors, Word2Vec and the Skip-gram model (Chapters 1–5): cs224d.stanford.edu
- Skipgram & CBOW Videos — Word2Vec architectures, Softmax, and Negative Sampling: [Watch on YouTube](#)
- Jurafsky & Martin — SLP Chapter 13 — RNNs, vanishing gradients, LSTMs, Seq2Seq, and Attention: web.stanford.edu

Additional Pre-reads (for those who wish to explore further)

For a deeper understanding, participants may refer to the following foundational resources on:

- Backpropagation — Step-by-step mathematical derivation: towardsdatascience.com
- Backpropagation Through Time — Comprehensive research paper on BPTT and gradient challenges: arxiv.org/abs/1802.01528
- Book: Understanding Deep Learning, Simon J.D. Prince — MIT Press (2023): udlbook.github.io
- Book: Deep Learning, Goodfellow, Bengio & Courville — CNN foundations and optimization: deeplearningbook.org

Post-read Materials

After the lecture, participants are encouraged to refer to the following resources for a more detailed understanding of the topics discussed.

Introduction to Deep Learning — Andrew Ng: [Watch on YouTube](#)

- Cost Function & Gradient Descent (Videos 5, 6)
- Overfitting, Underfitting & Model Evaluation (Videos 7, 8)

Convolutional Neural Networks — Andrew Ng: [Watch on YouTube](#)

- CNN architectural foundations: filters, pooling, strides
- Operational mechanics of convolution
- Deep CNN architectures and training

Word2Vec — Efficient Estimation of Word Representations in Vector Space: arxiv.org/abs/1301.3781

- Theoretical underpinnings of word embeddings
- Skip-gram and CBOW operational mechanics
- Negative sampling and efficiency techniques

Research Papers — CNN Architectures

- [AlexNet — ImageNet Classification with Deep CNNs \(Krizhevsky et al., 2012\)](#)
- [ResNet — Deep Residual Learning for Image Recognition \(He et al., 2016\)](#)

Hands-on Tutorials —

- Practice Worksheet - [CNN Worksheet Exercise](#)
- PyTorch Image Classification (CIFAR-10): docs.pytorch.org
- PyTorch Sequence Modeling Tutorial: docs.pytorch.org